



STEM SMART Single Maths 25 - Sequences & Series >

Arithmetic Sequences and Series

A-level Maths Topic Summaries - Series

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P2

Fill in the blanks below to complete the notes on arithmetic sequences and series.

Part A

Arithmetic sequences

An example of an arithmetic sequence is

3 5 7 9 11 13 ...

In an arithmetic sequence there is a constant difference between successive terms. If is the first term in an arithmetic sequence, and is the common difference,

1st term: $u_1 = a$

2nd term: $u_2 = u_1 + d = a + d$

3rd term: $u_3 = u_2 + d = a + 2d$

...

The n th term in the sequence is given by the formula

$$u_n = a + \text{} d$$

Items:

Part B

Arithmetic series

An example of an arithmetic series is

$$3 + 5 + 7 + 9 + 11 + 13 + \dots$$

An arithmetic series is a sum of terms in an arithmetic sequence. The sum of the first n terms, S , is given by

$$S_n = u_1 + u_2 + u_3 + \dots + u_n$$

$$S_n = \frac{n}{2} \boxed{} a + \boxed{} d$$

If the value of the last term in the series is l , this formula can also be written

$$S_n = \frac{n}{2} a + \boxed{} l$$

Items:

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Geometric Sequences and Series

A-level Maths Topic Summaries - Series

Subject & topics: Maths | Algebra | Series

Stage & difficulty: A Level P2

Fill in the blanks below to complete the notes on geometric sequences and series.

Part A

Geometric sequences

An example of a geometric sequence is

3 6 12 24 48 96 ...

In a geometric sequence there is a constant ratio between successive terms. If is the first term in an geometric sequence, and is the common ratio,

1st term: $u_1 = a$ **2nd term:** $u_2 = u_1 \times r = ar$ **3rd term:** $u_3 = u_2 \times r = ar^2$

...

The n th term in the sequence is given by the formula

$$u_n = a \text{ }$$

Items:



Question

Sequences and Series 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A sequence of terms $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 2 \text{ and } u_{n+1} = 1 - u_n$$

$$\text{for } n \geq 1$$

Part A Values

Give the values of u_2 , u_3 and u_4 .

$u_2 =$

$u_3 =$

$u_4 =$

Part B**Behaviour**

Describe the behaviour of the sequence.

- ☐ The sequence is periodic, with a period of two. It alternates between values of 2 and -1 .
- ☐ The sequence is periodic, with a period of three. It cycles through values of 2, -1 and 1.
- ☐ The sequence is periodic, with a period of four. The first two values that repeat are 2 and -1 .
- ☐ It is a geometric sequence, with first term 2 and constant ratio $-\frac{1}{2}$.

Part C**Sum**

Find $\sum_{n=1}^{100} u_n$.

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Arithmetic Progression 1

Pre-Uni Maths for Sciences C2.2

Subject & topics: Maths | Algebra | Series **Stage & difficulty:** A Level P2

In an arithmetic progression, the fifth term is 32 and the tenth term is 57.

Part A

First term

Find the first term, a .

Part B

Common difference

Find the common difference, d .

Part C

Sum of first 70 terms

Hence, find the sum of the first 70 terms.

Created for isaacphysics.org by Julia Riley.

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Arithmetic Series 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

The tenth term of an arithmetic progression is equal to twice the fourth term. The twentieth term of the progression is 44.

Part A

First Term

Find the first term.

Part B

Common Difference

Find the common difference.

Part C

Sum of the Series

Find the sum of the first 50 terms.

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Geometric Series 1ii

Subject & topics: Maths Stage & difficulty: A Level P2

Records are kept of the number of copies of a certain book that are sold each week. In the first week after publication, 3000 copies were sold, and in the second week 2400 copies were sold. The publisher forecasts future sales by assuming that the number of copies sold each week will form a geometric progression with first two terms 3000 and 2400. Calculate (to the nearest number of whole books) the publisher's forecasts for:

Part A

20th Week

The number of copies that will be sold in the 20th week after publication.

Part B

Total copies sold in 20 weeks

The total number of copies sold during the first 20 weeks after publication.

Part C

Total sold copies

The total number of copies that will ever be sold.

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Geometric Series 2ii

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Geometric progression 1 - r

In a geometric progression, the sum to infinity is four times the first term.

Find the common ratio.

Part B

Geometric progression 1 - a

Given that the third term is 9, find the first term.

Part C

Geometric progression 1 - sum

Find the sum of the first twenty terms. Give your answer to 3 sf.

Part D

Geometric progression 2

In another geometric progression, the first term is 6 and the sum to infinity is 10.

Find the common ratio.

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Geometric Series 4ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

In a geometric progression, the first term is 5 and the second term is 4.8.

Part A

Sum to Infinity

Find the sum to infinity.

Part B

Value of n

The sum of the first n terms is greater than 124. By showing that

$$0.96^n < 0.008$$

and using logarithms, calculate the smallest possible value of n .

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Arithmetic Series 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

In an arithmetic progression the first term is 5 and the common difference is 3. The n^{th} term of the progression is denoted by u_n .

Part A

Value of u_{20}

Find the value of u_{20} .

The following symbols may be useful: `u_20`

Part B

Sum

Find the value of $\sum_{n=10}^{20} u_n$.

Part C

Value of N

Find the value of N such that $\sum_{n=N}^{2N} u_n = 2750$.

The following symbols may be useful: N

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